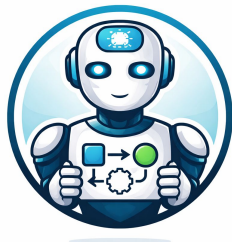


Agentic AI for Tech Professionals

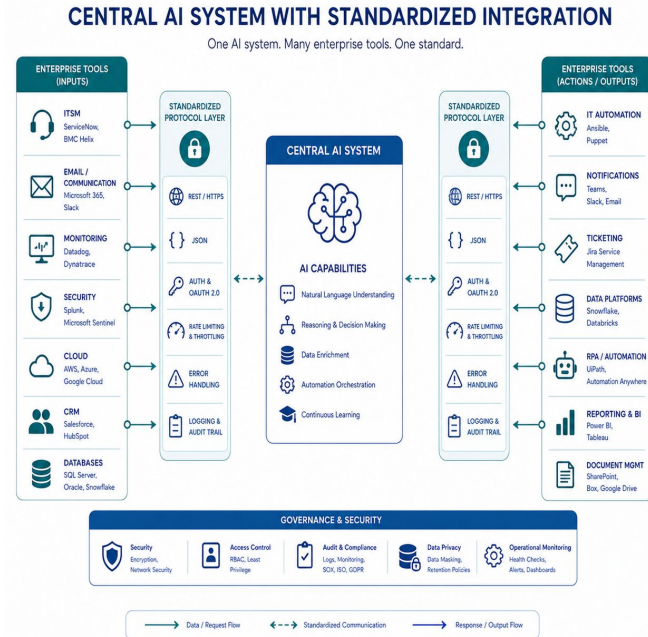
Model Context Protocol



What is MCP (Model Context Protocol)?

MCP standardizes how AI systems communicate with tools and data sources

- Common protocol for AI-tool interaction
- Standardizes requests and responses
- Simplifies enterprise integrations
- Enables reusable AI workflows
- Reduces custom integration logic



What is MCP (Model Context Protocol)?

As AI systems grow,
they need to interact with:

- APIs
- databases
- workflows
- internal systems

Without standards:

✗ every integration becomes custom

MCP introduces:

👉 a standardized
communication layer

Think of it like:

👉 USB for AI systems

One protocol.

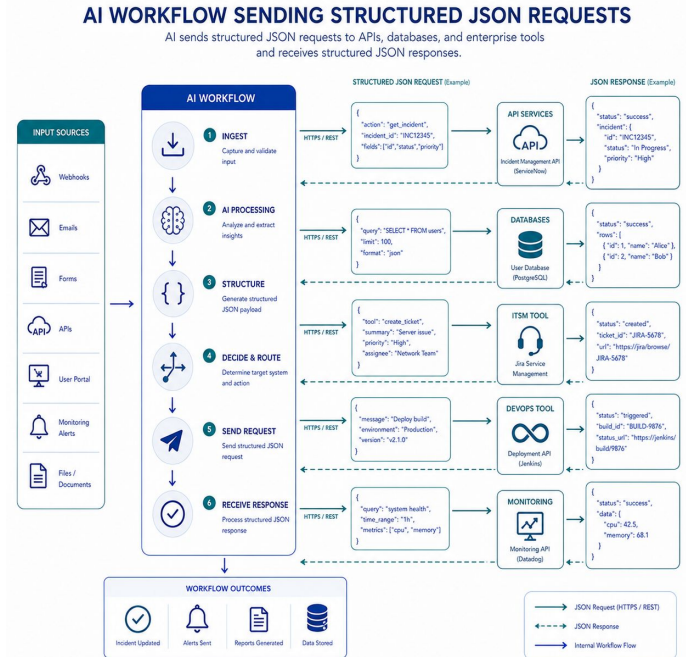
Many tools.

This simplifies enterprise AI
architecture significantly.

Standardizing AI-to-Tool Communication

MCP creates predictable communication between AI and enterprise systems

- AI sends structured requests
- Tools return structured responses
- Consistent interface across systems
- Easier orchestration and automation
- Supports scalable enterprise AI



Standardizing AI-to-Tool Communication (2)

Without MCP:

every tool works differently.

One API:

- different authentication
- different response format
- different logic

With MCP:

AI communicates consistently.

Standardizing AI-to-Tool Communication (3)

Example:

AI asks:

👉 “Retrieve incident history”

Tool responds:

👉 standardized JSON

This makes workflows:



reusable



scalable

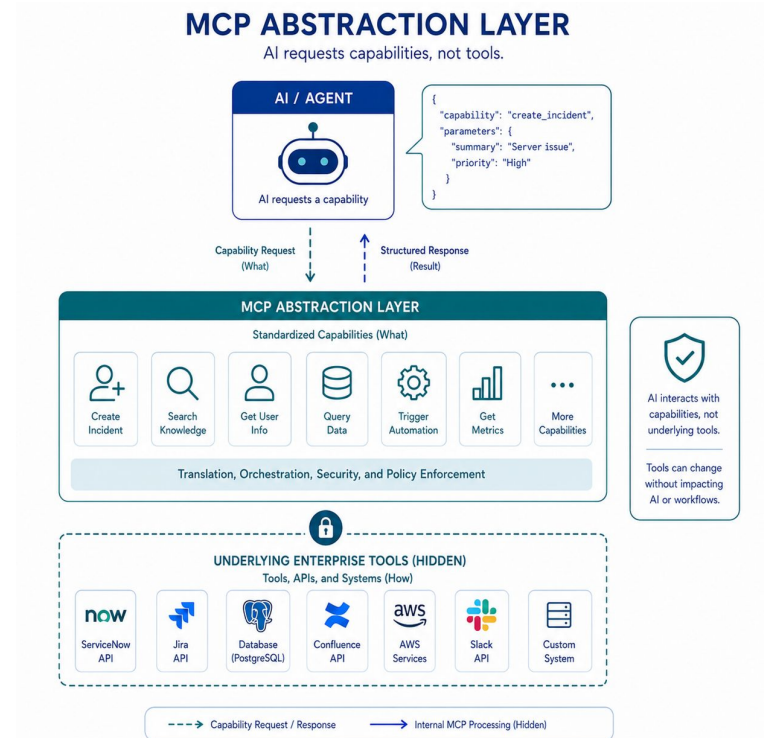


easier to maintain

Tool Abstraction

AI interacts with capabilities instead of tool-specific implementations

- AI focuses on task, not tool details
- Tools become interchangeable
- Simplifies workflow design
- Reduces dependency complexity
- Enables modular enterprise systems



Tool Abstraction (2)

Tool abstraction is extremely important.

Instead of AI knowing:

- PostgreSQL
- Google Sheets
- internal API details

AI only knows:

- 👉 “store incident”
- 👉 “retrieve SOP”
- 👉 “send notification”

The MCP layer handles:

- implementation
- authentication
- tool selection

This creates:

👉 modular enterprise AI systems

Enterprise Integration with MCP

MCP enables AI workflows to connect across enterprise systems

- Connect APIs and databases
- Integrate multiple workflows
- Standardize enterprise automation
- Improve interoperability
- Enable large-scale orchestration

Enterprise Integration with MCP (2)

Enterprise environments contain many systems:

- ERP
- ticketing systems
- databases
- workflow engines
- monitoring platforms

MCP helps unify them.

Enterprise Integration with MCP (3)

Example:

AI workflow:

- retrieves incident history
- queries SOP database
- creates maintenance ticket
- logs audit record

All through:

👉 one standardized interaction model

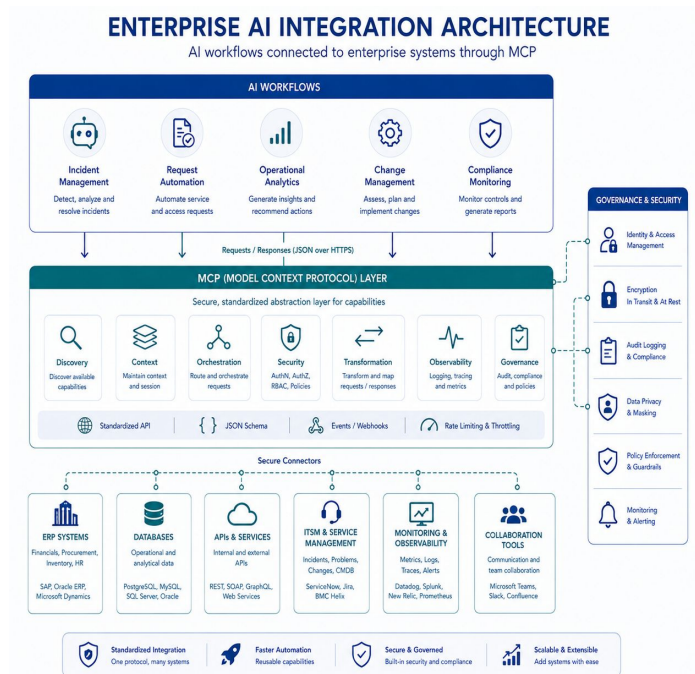
This is critical for:

👉 scalable enterprise AI adoption.

MCP in Manufacturing Incident Workflow

MCP standardizes communication across the entire AI incident ecosystem

- AI analyzes incident
- MCP connects enterprise tools
- Systems exchange structured data
- Workflow orchestration becomes modular
- Governance and monitoring improve



MCP in Manufacturing Incident Workflow (2)

Now the workflow becomes truly enterprise-grade.

Example:

Incident received.

AI workflow:

1. Retrieves SOP
2. Queries incident database
3. Creates maintenance ticket
4. Sends notification
5. Logs audit trail

MCP standardizes all these interactions.

This transforms isolated workflows into:

👉 connected enterprise AI ecosystems.